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**VIRILE AGITUR**



**Student Number**

**Knox Grammar School**

**2009**

**Trial Higher School Certificate  
Examination**

## Mathematics Extension 2

### General Instructions

- Reading time – 5 minutes
- Working time - 3 hours
- Write using blue or black pen only
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

### Subject Teachers

Mr R. Maxwell  
Mrs M. Wassef

**This paper MUST NOT be removed from the examination room**

**Number of Students in Course: 40**

**Number of Writing Booklets Per Student  
(Eight Page) 8**

### Total Marks – 120

- Attempt Questions 1 – 8
- Answer each question in a separate writing booklet
- All questions are of equal value

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Total marks – 120

Attempt Questions 1–8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks)

Marks

(a) Find  $\int \frac{dx}{\sqrt{16x^2 - 1}}$

2

(b) Evaluate  $\int_1^e x \ln x \, dx$

3

(c) (i) Find real numbers  $a$  and  $b$  such that

2

$$\frac{5x^2 + x + 8}{(x+1)(x^2+3)} \equiv \frac{a}{x+1} + \frac{bx-1}{x^2+3}$$

(ii) Hence find  $\int \frac{5x^2 + x + 8}{(x+1)(x^2+3)} \, dx$

2

(d) Find  $\int \tan^3 x \, dx$

2

(e) Using a suitable substitution, or otherwise, evaluate:

4

$$\int_0^2 \frac{x^2}{\sqrt{4-x^2}} \, dx$$

End of Question 1

Question 2 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Let  $\alpha = 1 - \sqrt{3}i$ .

(i) Find the exact value of  $|\alpha|$  and  $\arg \alpha$ .

2

(ii) Hence express  $(1 - \sqrt{3}i)^{10}$  in modulus-argument form.

1

(b) Express  $\sqrt{7 - 24i}$  in the form  $a + ib$ , where  $a$  and  $b$  are real.

3

(c) Sketch the region in the complex plane where the two inequalities  $0 \leq \arg(z) \leq \frac{3\pi}{4}$  and  $|z - 2i| \geq |z|$  both hold.

3

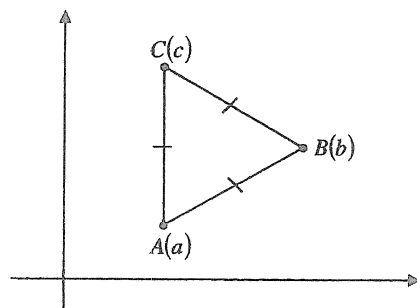
(d) Sketch the locus of  $z$  satisfying  $|z - 3| + |z + 3| = 10$ . Show any intercepts with the axes.

3

Question 2 continues on page 4

Question 2 (continued)

(e)



The points  $A$ ,  $B$  and  $C$  on the Argand diagram represent the complex numbers  $a$ ,  $b$  and  $c$  respectively.  $\triangle ABC$  is equilateral.

Let  $w = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$ .

(i) Show that  $\frac{a-b}{c-b} = w$ .

1

(ii) By writing another similar expression for  $w$ , prove that

2

$$a^2 + b^2 + c^2 = ab + bc + ca$$

End of Question 2

Question 3 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) The equation  $x^3 + 3x^2 - 5x - 2 = 0$  has roots  $\alpha$ ,  $\beta$  and  $\gamma$ .

2

Find a cubic equation with integer coefficients whose roots are  $\frac{2}{\alpha}$ ,  $\frac{2}{\beta}$  and  $\frac{2}{\gamma}$ .

(b) Consider the curve  $x^2 + y^2 + xy = 3$ .

(i) Show that  $\frac{dy}{dx} = -\left(\frac{2x+y}{x+2y}\right)$ .

1

(ii) Hence find the coordinates of any stationary points.

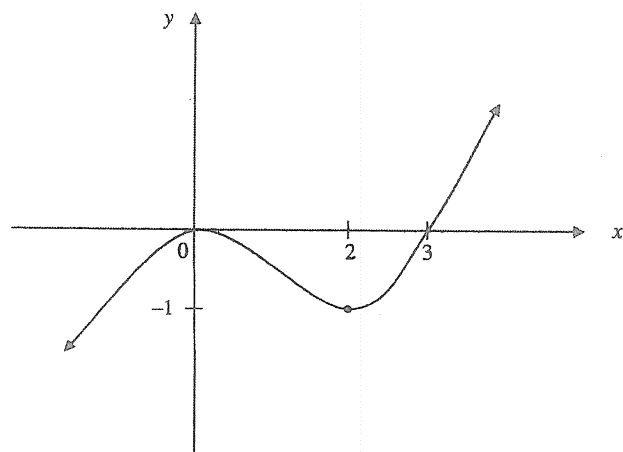
2

Question 3 continues on page 6

Question 3 (continued)

Marks

- (c) The diagram shows the graph of  $y = f(x)$  where  $f(x) = \frac{1}{4}x^2(x-3)$ .



On the answer page provided, draw separate sketches of the graphs of the following:

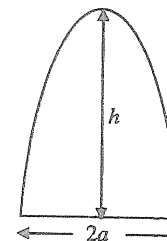
- |       |                            |   |
|-------|----------------------------|---|
| (i)   | $y = \frac{1}{4}x^2  x-3 $ | 1 |
| (ii)  | $y = \frac{1}{f(x)}$       | 1 |
| (iii) | $y^2 = -f(x)$              | 2 |
| (iv)  | $y = \tan^{-1}(f(x))$      | 2 |

Question 3 continues on page 7

Question 3 (continued)

Marks

- (d) (i)

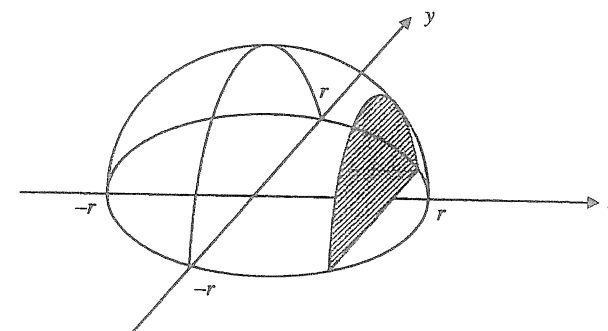


1

A parabolic segment has height  $h$  and width  $2a$ .

Use Simpson's Rule with three function values, to show that the exact area of this segment is  $\frac{4ah}{3}$ .

- (ii)



3

The base of a solid is the region in the  $xy$  plane enclosed by the circle  $x^2 + y^2 = r^2$ .

Each cross-section perpendicular to the  $x$ -axis is a parabolic segment with height one half its width.

Show that the volume of the solid is  $\frac{16r^3}{9}$  units<sup>3</sup>.

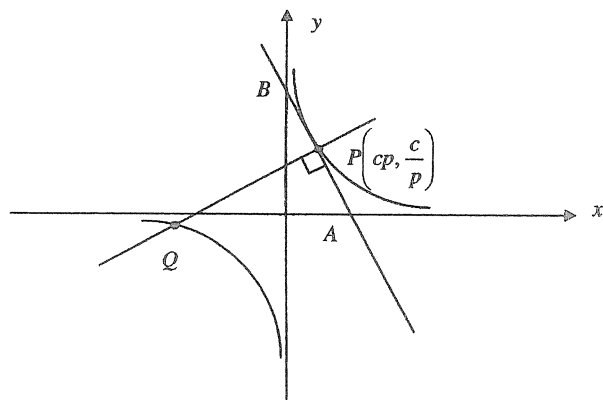
End of Question 3

**Question 4 (15 marks)** Use a SEPARATE writing booklet.

Marks

- (a) The point  $P\left(cp, \frac{c}{p}\right)$  is a point on the hyperbola  $xy = c^2$ .

The tangent to the hyperbola at  $P$  intersects the  $x$  and  $y$  axes at  $A$  and  $B$  respectively and the normal to the hyperbola at  $P$  intersects the second branch at  $Q$ .



- (i) Show that the equation of the normal at  $P$  is  $py - c = p^3(x - cp)$ . 2

- (ii) Show that the  $x$  coordinates of  $P$  and  $Q$  satisfy the equation 2

$$x^2 - c\left(p - \frac{1}{p^3}\right)x - \frac{c^2}{p^2} = 0$$

and hence find the coordinates of  $Q$ .

- (iii) Given the distance  $AB = 2c\sqrt{p^2 + \frac{1}{p^2}}$ , show that the 2

$$\text{area of } \triangle ABQ = c^2\left(p^2 + \frac{1}{p^2}\right)^2.$$

- (iv) Find the minimum area of  $\triangle ABQ$ . 1

(You may use the inequality  $\frac{a}{b} + \frac{b}{a} \geq 2$  for  $a, b > 0$ .)

**Question 4 continues on page 9****Question 4 (Continued)**

Marks

- (b) An ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  has foci  $S$  and  $S'$  where  $b^2 = a^2(1 - e^2)$  where  $0 < e < 1$  is its eccentricity.

- (i) Show that the equation of the normal to the ellipse at the point  $P(x_1, y_1)$  is 2

$$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2.$$

- (ii) Show that  $N$ , the  $x$ -intercept of the normal has coordinates  $N(e^2x_1, 0)$ . 1

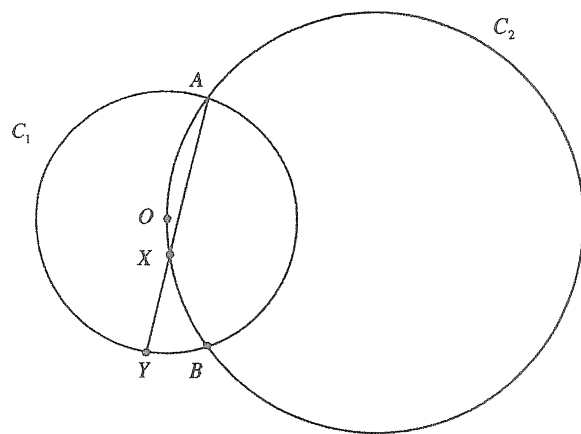
- (iii) Use the focus/directrix definition of a conic to prove that  $\frac{SN}{S'N} = \frac{SP}{S'P}$  2

**Question 4 continues on page 10**

Question 4 (continued)

Marks

(c)



Two circles  $C_1$  and  $C_2$  intersect at  $A$  and  $B$ .  $C_2$  passes through  $O$ , the centre of  $C_1$ .  $X$  lies on the arc  $AOB$  and  $AX$  intersects  $C_1$  again at  $Y$ .

(i) State why  $\angle AOB = 2 \times \angle AYB$ .

1

(ii) Prove that  $XY = XB$ .

3

Question 5 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) (i) Show that if  $\alpha$  is a double root of  $f(x) = 0$  then  $f(\alpha) = f'(\alpha) = 0$ .

2

(ii) Find all roots of the equation  $2x^3 - 5x^2 - 4x + 12 = 0$  given that two of the roots are equal.

3

(b) (i) By drawing a diagram, or otherwise, find the solutions of  $z^5 = 1$ .

2

(ii) Show that  $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$ .

2

(iii) Hence find the exact value of  $\cos \frac{2\pi}{5}$ .

2

(c) 11 persons gather to play basketball by forming 2 teams of 5 to play each other. The remaining person acts as a referee.

(i) In how many ways can the teams be formed?

2

(ii) If two particular persons are not to be in the same team, how many ways are there then to choose the teams?

2

**Question 6** (15 marks) Use a SEPARATE writing booklet.

Marks

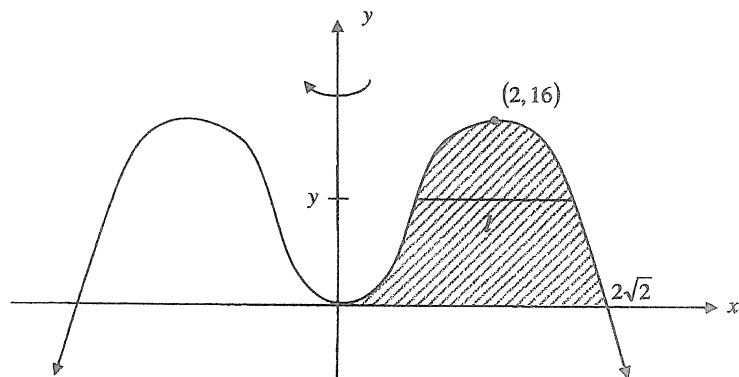
- (a) The sequence
- $\{a_n\}$
- is given by:

3

$$a_1 = 2, a_2 = \frac{3}{2} \text{ and } (n+1)a_{n+1} = a_{n-1} - (n-2)a_n \text{ for } n > 1.$$

Prove by induction that for  $n \geq 1$ ,  $a_n = \frac{n+1}{n!}$

- (b) The region bound by the curve  $y = 8x^2 - x^4$  and the  $x$  axis in the first quadrant is rotated about the  $y$  axis to form a solid. When the region is rotated, the horizontal line segment  $l$  at height  $y$  sweeps out an annulus.



- (i) Show that the area of the annulus at height  $y$  is given by  $2\pi\sqrt{16-y}$ .

3

- (ii) Find the volume of the solid.

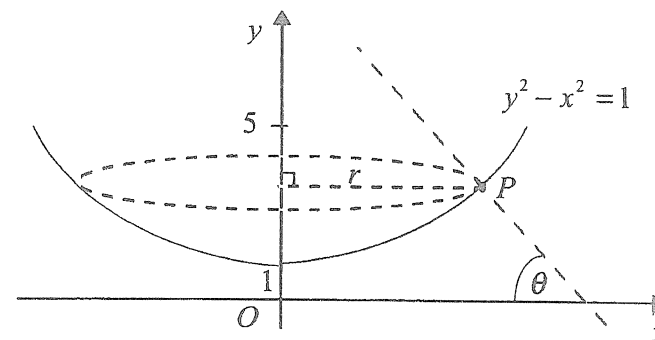
2

Question 6 continues on page 13

Question 6 (continued)

Marks

- (c)



A bowl is formed by rotating the hyperbola  $y^2 - x^2 = 1$  for  $1 \leq y \leq 5$  through  $180^\circ$  about the  $y$ -axis. Sometime later, a particle  $P$  of mass  $m$  moves around the inner surface of the bowl in a horizontal circle with constant angular velocity  $\omega$ .

- (i) Show that if the radius of the circle in which  $P$  moves is  $r$ , then the normal to the surface at  $P$  makes an angle  $\theta$  with the horizontal where  $\tan \theta = \frac{\sqrt{1+r^2}}{r}$ .
- (ii) Find expressions for the radius  $r$  of the circle of motion and the magnitude of the reaction force between the surface and the particle in terms of  $m, g$  and  $\omega$ .
- (iii) Find the values for  $\omega$  for which the described motion of  $P$  is possible.

3

2

3

End of Question 6

**Question 7 (15 Marks)** Use a SEPARATE writing booklet.

**Marks**

- (a) A particle of mass  $m$  is projected vertically upwards with an initial velocity of  $V$  in a medium where the resistance force  $R$  to the motion has a magnitude  $R = mkv$  where  $v$  is the velocity of the particle after the initial projection.
- (i) Show that the maximum height  $h$  of the particle is given by 3  

$$h = \frac{g}{k^2} \left\{ \frac{k}{g} V - \ln \left( 1 + \frac{k}{g} V \right) \right\}$$
- (ii) Find an expression for the time  $T$  of particle to reach its maximum height in terms of  $V$ ,  $k$  and  $g$ . 2
- (iii) After reaching its maximum height the particle returns vertically downwards towards its projection point in the same medium. Show that the downward velocity is given by  $v = \frac{g}{k} (1 - e^{-kt})$  where  $t$  is the time of the downward motion and give the terminal velocity of the particle. 3
- (iv) The speed of upward projection is double the terminal velocity and the particle's downward displacement  $y$  from its maximum height is given by the equation  $-\frac{k^2}{g} y = \frac{k}{g} v + \ln \left( 1 - \frac{k}{g} v \right)$ . (Do NOT prove this result).  
 Show that the velocity of the particle on return to its projection point is given by  $\frac{kv}{g} + 2 - \ln 3 + \ln \left( 1 - \frac{kv}{g} \right) = 0$ . If a root to this equation for  $\frac{kv}{g}$  lies near 0.81, use one application of "Newton's Method" to find a better approximation correct to two decimal places and deduce the percentage of the terminal velocity that the particle has acquired on return to its projection point. 3
- (v) Hence find the ratio of the time taken to reach maximum height to the time to fall from maximum height to the point of projection. 1
- (b) When an unbiased coin is tossed  $2n$  times, the probability of observing  $k$  heads and  $2n - k$  tails is given by  $P_k = {}^{2n}C_k \left( \frac{1}{2} \right)^k \left( \frac{1}{2} \right)^{2n-k}$ . 2
- (i) Show that the most likely outcome is  $k = n$  1
- (ii) Show that  $P_n = \frac{(2n)!}{2^{2n} (n!)^2}$

**End of Question 7**

**Question 8 (15 Marks)** Use a SEPARATE writing booklet.

**Marks**

- (a) Given  $I_n = \int_1^a (1 - \ln x)^n dx$  for  $n = 0, 1, 2, \dots$
- (i) Show that  $I_n = -1 + nI_{n-1}$  for  $n = 1, 2, 3, \dots$  2
- (ii) Hence evaluate  $\int_1^e (1 - \ln x)^3 dx$  2
- (iii) Show that  $\frac{I_n}{n!} = e - \sum_{r=0}^n \frac{1}{r!}$  for  $n = 1, 2, 3, \dots$  2
- (iv) Show that  $0 \leq I_n \leq e - 1$  1
- (v) Deduce that  $\lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{1}{r!} = e$  1
- (b)  $\phi(x)$  and  $\psi(x)$  are continuous and bounded functions.
- (i) By considering  $\int_0^a \{ \lambda \phi(x) + \psi(x) \}^2 dx$  for  $a > 0$  as a quadratic function in  $\lambda$ , show that  $\left\{ \int_0^a \phi(x) \psi(x) dx \right\}^2 \leq \int_0^a \{ \phi(x) \}^2 dx \times \int_0^a \{ \psi(x) \}^2 dx$ . 4
- (ii) Hence show that  $\left\{ \int_0^1 \phi(x) dx \right\}^2 \leq \int_0^1 \{ \phi(x) \}^2 dx$ . 2
- (iii) Deduce that  $\left\{ \int_0^1 \phi(x) dx \right\}^4 \leq \int_0^1 \{ \phi(x) \}^4 dx$ . 1

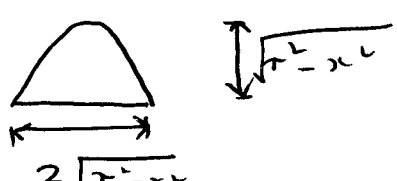
**End of Paper**

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| 2. MAX | 6. MAX |
| 3. HAR | 7. WAM |
| 4. DIN | 8. MAX |

**2009 Year 12 Mathematics Extension 2 Task 4 Trial HSC SOLUTIONS**

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Comments | Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Comments |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <p><u>Question 1</u></p> <p>a) <math>\int \frac{dx}{\sqrt{16x^2-1}} = \frac{1}{4} \int \frac{dx}{\sqrt{x^2 - (\frac{1}{4})^2}}</math></p> $= \frac{1}{4} \ln(x + \sqrt{x^2 - \frac{1}{4}}) + C$ $= \frac{1}{4} \ln(4x + \sqrt{16x^2-1}) + C$ <p>✓ coeff of <math>\frac{1}{4}</math></p> <p>✓ answer or equiv.</p> <p>b) <math>\int_1^e x \ln x</math>     <math>u = \ln x \quad v' = x</math><br/> <math>u' = \frac{1}{x} \quad v = \frac{x^2}{2}</math></p> $= \left[ \frac{x^2 \ln x}{2} \right]_1^e - \int_1^e \frac{x}{2} dx$ $= \frac{e^2}{2} - \left[ \frac{x^2}{4} \right]_1^e$ $= \frac{e^2}{4} + \frac{1}{4}$ <p>c) i) <math>a=3</math> ✓ <math>b=2</math> ✓</p> <p>ii) <math>\int \frac{5x^2+x+8}{(x+1)(x+3)} dx = \int \frac{3}{x+1} + \frac{2x-1}{x^2+3} dx</math></p> $= 3 \ln x+1  + \ln(x^2+3) - \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{x}{\sqrt{3}}\right) + C$ <p>d) <math>\int \tan^3 x dx = \int \tan x (\sec^2 x - 1) dx</math></p> $= \int \sec^2 x \tan x - \frac{\sin x}{\cos x} dx$ $= \frac{1}{2} \tan^2 x + \ln \cos x  + C$ <p>e) <math>\int_0^2 \frac{x^2}{\sqrt{4-x^2}} dx</math>     let <math>x = 2 \sin \theta</math><br/> <math>dx = 2 \cos \theta d\theta</math><br/> <math>x=0, \theta=0</math><br/> <math>x=2, \theta=\frac{\pi}{2}</math></p> $= \int_0^{\frac{\pi}{2}} \frac{4 \sin^2 \theta}{2 \cos \theta} \cdot 2 \cos \theta d\theta$ $= \int_0^{\frac{\pi}{2}} 4 \sin^2 \theta d\theta = \int_0^{\frac{\pi}{2}} 2(1 - \cos 2\theta) d\theta$ $= [2\theta - \sin 2\theta]_0^{\frac{\pi}{2}} = \pi$ |          | <p><u>Question 2</u></p> <p>a) i) <math> z =2 \quad \arg(z) = -\frac{\pi}{3}</math></p> <p>ii) <math>(1-3i)^{10} = (2 \operatorname{cis}(-\frac{\pi}{3}))^{10}</math></p> $= 2^{10} \operatorname{cis}(-\frac{10\pi}{3})$ $= 2^{10} \operatorname{cis}(\frac{2\pi}{3}) \quad \checkmark$ <p>b) let <math>(a+bi)^2 = 7-24i</math></p> $\therefore a^2 - b^2 + 2abi = 7 - 24i \quad \checkmark$ <p>ie, <math>a^2 - b^2 = 7 \quad 2ab = -24</math><br/> <math>ab = -12</math></p> <p>by inspection <math>a = \pm 4 \quad a, b \in \mathbb{R}</math><br/> <math>b = \mp 3</math></p> $\therefore \sqrt{7-24i} = 4-3i, -4+3i$ <p>c)</p> <p>✓ <math>0 \leq \arg(z) \leq \frac{\pi}{4}</math><br/> <math>\sqrt{17-2i} =  z </math><br/> <math>\checkmark</math> region deduced &amp; origin included.</p> <p>d)</p> <p>✓ ellipse with foci <math>(\pm 3, 0)</math><br/> <math>\checkmark</math> x-intercepts <math>\pm 5</math><br/> <math>\checkmark</math> y-intercepts <math>\pm 4</math></p> <p>e) i) <math>\vec{BA} = (a-b)</math>, <math>\vec{BC} = (c-b)</math></p> <p>Since <math>\vec{BA} = \vec{BC} \times \operatorname{cis}(\frac{\pi}{3})</math> ✓</p> $(a-b) = (c-b) \times \omega$ $\therefore \omega = \frac{a-b}{c-b}$ <p>ii) Similarly <math>\frac{c-a}{b-a} = \omega</math> ✓</p> $\therefore \frac{a-b}{c-b} = \frac{c-a}{b-a} \Rightarrow (a-b)(b-a) = (c-a)(c-b)$ $ab - a^2 - b^2 + ab = c^2 - cb - ac + ab$ $\therefore a^2 + b^2 + c^2 = ab + bc + ca$ <p>✓ progress to.</p> |          |

**2009 Year 12 Mathematics Extension 2 Task 4 Trial HSC SOLUTIONS**

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Comments | Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Comments |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <p><u>Question 3</u></p> <p>a) let <math>y = \frac{z}{x}</math><br/> <math>(\frac{z}{x})^3 + 3(\frac{z}{x})^2 - 5(\frac{z}{x}) - 2 = 0</math> ✓<br/> <math>\Rightarrow y^3 + 5y^2 - 6y + 4 = 0</math> ✓</p> <p>b) <math>x^2 + y^2 + xy = 3</math></p> <p>i) <math>2x + 2y \cdot \frac{dy}{dx} + (x \frac{dy}{dx} + y) = 0</math> ✓<br/> <math>(x + 2y) \frac{dy}{dx} = -2x - y</math><br/> <math>\frac{dy}{dx} = \frac{-(2x + y)}{x + 2y}</math></p> <p>ii) For stat. pts <math>\frac{dy}{dx} = 0</math><br/> <math>y = -2x</math><br/> <math>\therefore x^2 + (-2x)^2 + x(-2x) = 3</math><br/> <math>3x^2 = 3</math><br/> <math>x = \pm 1</math><br/> <math>y = \mp 2</math><br/> <math>\therefore</math> stat pts <math>(1, -2)</math> and <math>(-1, 2)</math> ✓</p> <p>c) see separate pages</p> |          | <p>d) i) <math>A = \frac{h}{3} [y_1 + 4y_2 + y_3]</math><br/> <math>= \frac{a}{3} [0 + 4h + 0]</math><br/> <math>= \frac{4ah}{3}</math> ✓</p> <p>ii) Cross-section:</p>  <p><math>\therefore \text{Area} = \frac{4ah}{3} = \frac{4}{3}(r^2 - x^2)</math></p> <p><math>V = 2 \int_0^r \frac{4}{3}(r^2 - x^2) dx</math> ✓<br/> <math>= \frac{8}{3} \left[ r^2 x - \frac{x^3}{3} \right]_0^r</math> ✓<br/> <math>= \frac{16r^3}{9} \text{ u}^3</math> ✓</p> |          |

HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION  
Mathematics Extension 2

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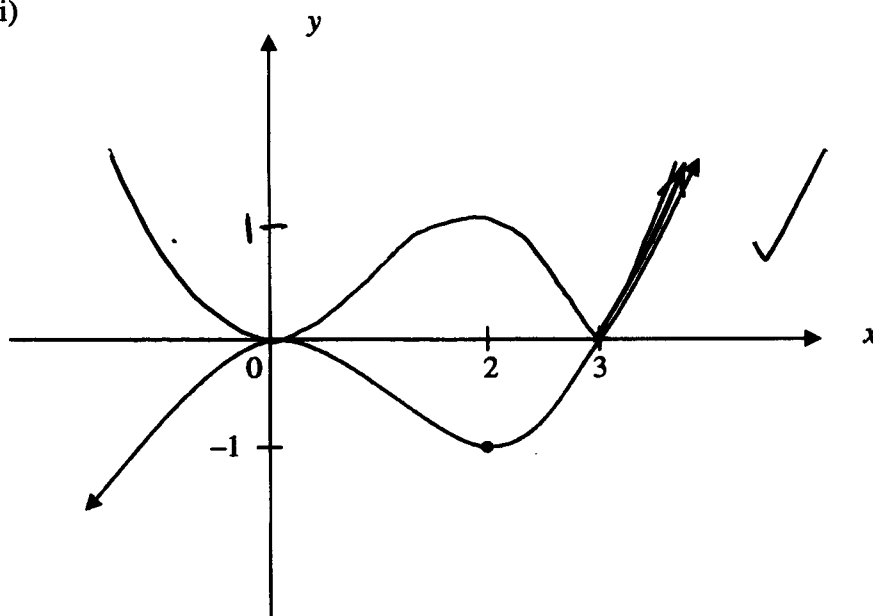
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Questions 3 (c)

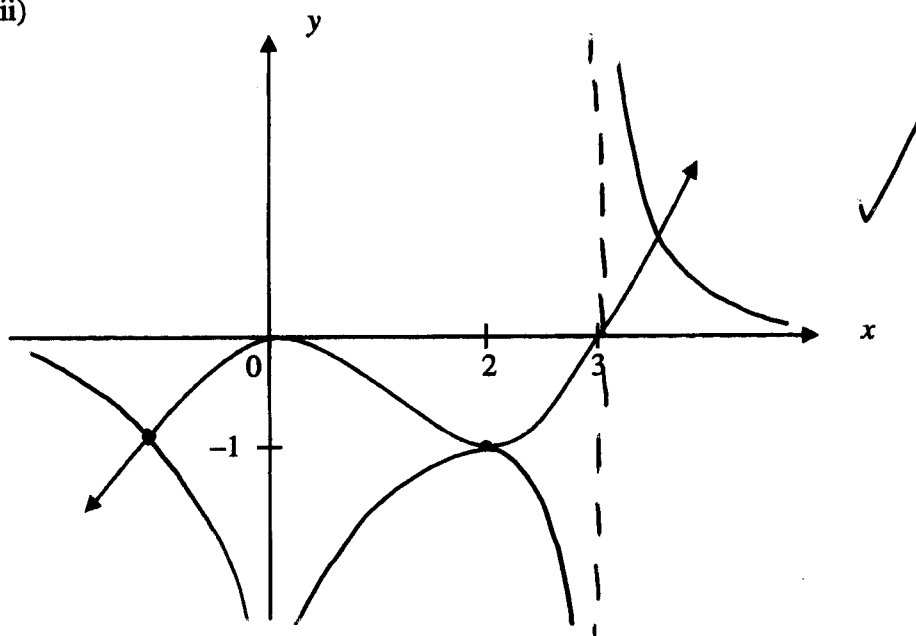
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Student Number

(i)



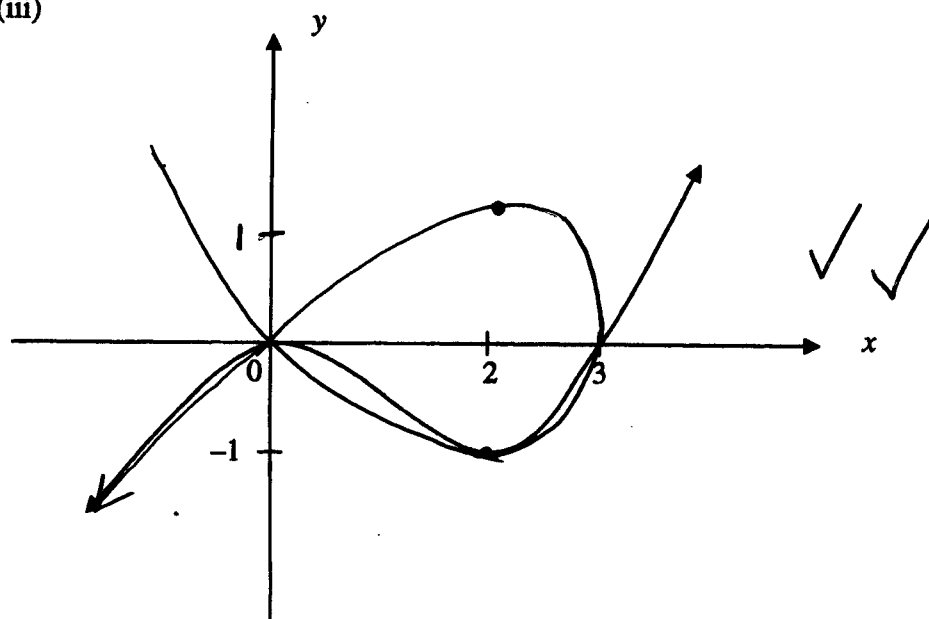
(ii)



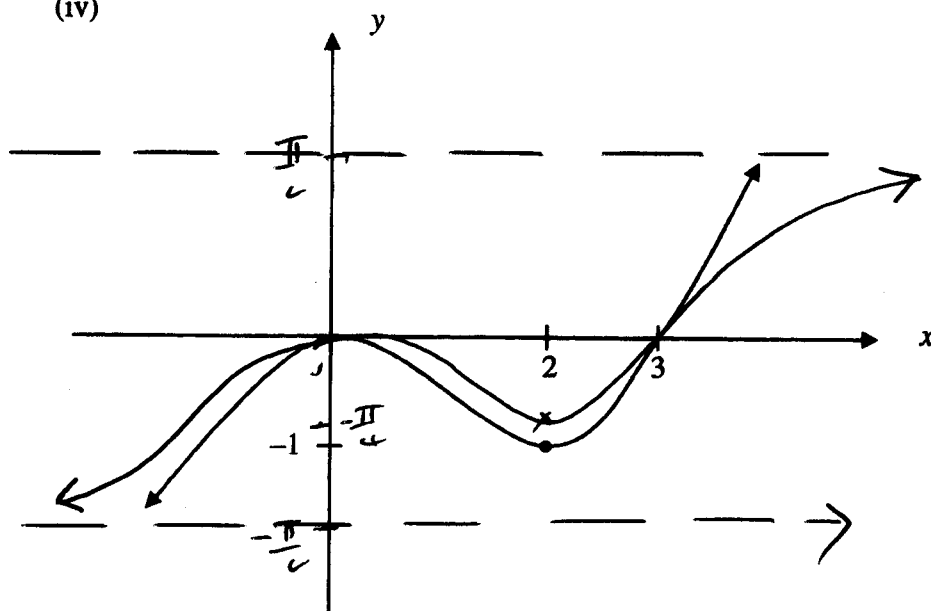
Question 3(c) continues on next page

Question 3 (continued)

(c) (iii)



(iv)



### Question 4

a) i)  $xy = c^2$   
 $x \cdot \frac{dy}{dx} + y = 0$   
 $\frac{dy}{dx} = -\frac{y}{x}$

at  $P(cp, \frac{c}{p})$   $\frac{dy}{dx} = -\frac{\frac{c}{p}}{cp}$

$\therefore m_T = -\frac{1}{p^2}$

$m_T = p^2$  ✓

$\therefore y - \frac{c}{p} = p^2(x - cp)$

$py - c = p^3(x - cp)$  ✓

ii) Solving  $y = \frac{c^2}{x}$  — ①

$py - c = p^3(x - cp)$  — ②

$p \cdot \frac{c^2}{x} - c = p^3(x - cp)$

$pc^2 - cx = p^3x^2 - cp^4x$

$\div p^3:$   $\frac{c^2}{p^2} - \frac{c}{p^2}x = x^2 - cp^2x$

$x^2 - c(p - \frac{1}{p^2})x - \frac{c^2}{p^2} = 0$

Let  $x$  coord of  $Q$  be  $\alpha$

$\therefore \alpha + cp = c(p - \frac{1}{p^2})$

$\therefore \alpha = -\frac{c}{p^3}$

sum of roots

$\therefore Q(-\frac{c}{p^3}, -cp^3)$  ✓

✓ - correct  
progress  
to  
equation

$$\text{iii)} \quad AB = 2c \sqrt{\rho^2 + \frac{1}{\rho^2}} \quad (\text{given})$$

$$\therefore PQ = \sqrt{\left(c\rho + \frac{c}{\rho^3}\right)^2 + \left(\frac{c}{\rho} + c\rho^3\right)^2}$$

$$= c \sqrt{\rho^2 + \frac{2}{\rho^2} + \frac{1}{\rho^6} + \frac{1}{\rho^2} + 2\rho^2 + \rho^6}$$

$$= c \sqrt{\rho^6 + 3\rho^2 + \frac{3}{\rho^2} + \frac{1}{\rho^6}}$$

$$= c \sqrt{\left(\rho^2 + \frac{1}{\rho^2}\right)^3} \quad \checkmark$$

$$\text{Area } \triangle ABQ = \frac{1}{2} \times AB \times PQ$$

$$= \frac{1}{2} \times 2c \sqrt{\rho^2 + \frac{1}{\rho^2}} \times c \sqrt{\left(\rho^2 + \frac{1}{\rho^2}\right)^3}$$

$$= c^2 \left(\rho^2 + \frac{1}{\rho^2}\right)^2 \quad \checkmark$$

$$\text{iv)} \quad \text{if } \frac{a}{b} + \frac{b}{a} \geq 2$$

$$\rho^2 + \frac{1}{\rho^2} \geq 2$$

$$\therefore \text{Area } \triangle ABQ \geq c^2 \times 2^2$$

$$\geq 4c^2$$

$$\therefore \text{Min. Area is } 4c^2. \quad \checkmark$$

$$(b) \quad i) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\text{at } P(x_1, y_1) \quad \frac{dy}{dx} = -\frac{b^2 x_1}{a^2 y_1} \quad \checkmark$$

$$\therefore \text{gradient of normal} = +\frac{a^2 y_1}{b^2 x_1} \quad \checkmark$$

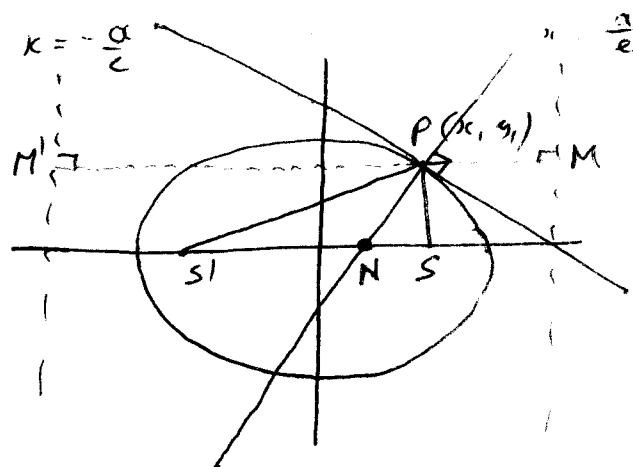
$$y - y_1 = \frac{a^2 y_1}{b^2 x_1} (x - x_1)$$

$$\frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2$$

$$ii) \quad \text{at } y=0 \quad \begin{aligned} a^2 x &= x_1 (a^2 - b^2) \\ a^2 x &= x_1 (a^2 - a^2(1 - e^2)) \\ a^2 x &= x_1 a^2 e^2 \\ \therefore x &= e^2 x_1 \quad \checkmark \end{aligned}$$

$$\therefore N(e^2 x_1, 0)$$

$$iii) \quad R.H.S = \frac{PS}{PS'} \\ = \frac{e PM}{e PM'} \\ = \frac{\frac{a}{e} - x_1}{\frac{a}{e} + x_1} \quad \checkmark \\ = \frac{a - ex_1}{a + ex_1}$$



$$L.H.S = \frac{SN}{S'N}$$

$$= \frac{ae - e^2 x_1}{ae - e^2 x_1} = \frac{e(a - ex_1)}{e(a - ex_1)}$$

$$\checkmark \therefore \frac{SN}{S'N} = \frac{PS}{S'P}$$

4(c) i) If  $\angle XYB = \alpha$   
 $\angle AOB = 2\alpha$  (L at centre is double L <sup>at circumference</sup> standing on same arc). ✓

$\angle AXB = \angle AOB = 2\alpha$  (L's in same segment are equal). ✓

$$\begin{aligned}\therefore \angle XBY &= \angle AXB - \angle XYB \\ &= 2\alpha - \alpha \\ &= \alpha\end{aligned}$$

(Ext L. of  $\Delta$  equals sum of 2 opp. int. L's) ✓

$\therefore XY = XB$  (sides opposite equal L's in a  $\Delta$  are equal)

### Question 5

(a) (i) If  $\alpha$  is a double root of  $f(x) = 0$ , then  $f(x)$  can be written:

$$f(x) = (x - \alpha)^2 \cdot Q(x)$$

$$f'(x) = 2(x - \alpha) \cdot Q(x) + (x - \alpha)^2 \cdot Q'(x)$$

$$= (x - \alpha) [2Q(x) + (x - \alpha) \cdot Q'(x)]$$

$$= (x - \alpha) \cdot \Phi(x)$$

$$f(\alpha) = (\alpha - \alpha)^2 \cdot Q(\alpha) \quad f'(\alpha) = (\alpha - \alpha) \cdot \Phi(\alpha)$$

$$= 0 \cdot Q(\alpha)$$

$$= 0 \cdot \Phi(\alpha)$$

$$= 0$$

$$\therefore f(\alpha) = f'(\alpha) = 0$$

(ii) let  $f(x) = 2x^3 - 5x^2 - 4x + 12$

$$\text{so } f'(x) = 6x^2 - 10x - 4 = 0$$

$$2x^2 - 5x - 2 = 0$$

$$(3x + 1)(x - 2) = 0$$

$$x = -\frac{1}{3}, 2$$

$$\text{and } f(2) = 0$$

$\therefore x = 2$  is the double root

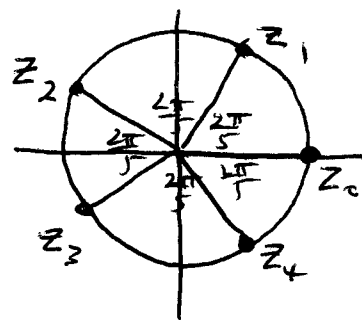
$$\text{so } 2 + 2 + \beta = \frac{5}{2}$$

$$\beta = -\frac{3}{2}$$

$\therefore$  Roots are  $2, 2, -\frac{3}{2}$

5b) i) 
$$\left. \begin{aligned} z_0 &= 1 \\ z_1 &= \text{cis } \frac{2\pi}{5} \\ z_2 &= \text{cis } \frac{4\pi}{5} \\ z_3 &= \text{cis } -\frac{4\pi}{5} \\ z_4 &= \text{cis } -\frac{2\pi}{5} \end{aligned} \right\}$$

✓ All roots



✓ diagram or otherwise

ii)

Sum of roots  $= -\frac{b}{a}$

$\therefore z_0 + z_1 + z_2 + z_3 + z_4 = -\frac{0}{1}$

✓  $z^5 - 1 = 0$

Noting  $\bar{z}_1 = z_4$   $\bar{z}_2 = z_3$

$1 + z_1 + \bar{z}_1 + z_2 + \bar{z}_2 = 0$

Note  $z + \bar{z} = 2\text{Re}(z)$

$\therefore 1 + 2\cos\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} = 0$

$\therefore \cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} = -\frac{1}{2}$

iii)

$\cos\frac{2\pi}{5} + 2\cos^2\frac{2\pi}{5} - 1 = -\frac{1}{2}$

$4\cos^2\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} + 1 = 0$

$\therefore \cos\frac{2\pi}{5} = -\frac{1 \pm \sqrt{5}}{4}$  but  $\cos\frac{2\pi}{5} > 0$

$\therefore \cos\frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{4}$

C i)  $\frac{{}^{11}C_5 \times {}^6C_5 \times {}^1C_1}{2} = 1386$

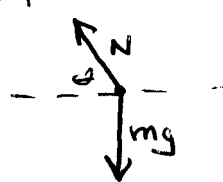
ii)  $9C_4 \times {}^5C_4 \times {}^1C_1 + \frac{{}^2C_1 \times {}^{10}C_5 \times {}^5C_5}{2}$   
 $= 630 + 252 = 882$

## 2009 Year 12 Mathematics Extension 2 Task 4 Trial HSC SOLUTIONS

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Comments                   | Suggested Solution (s)                                                                                                                                                                                                                                                                                                    | Comments |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <p><u>Question 6</u> a)</p> <p>When <math>n=1</math>, <math>\frac{1+1}{1!} = 2 = a_1</math></p> <p>When <math>n=2</math>, <math>\frac{2+1}{2!} = \frac{3}{2} = a_2</math></p> <p><math>\therefore</math> statement True for <math>n=1, 2</math></p> <p>Assume the statement true for <math>n=k-1</math> and <math>n=k</math></p> <p>i.e. <math>a_{k-1} = \frac{k}{(k-1)!}</math></p> <p><math>a_k = \frac{k+1}{k!}</math></p> <p>Let <math>n=k+1</math></p> <p>L.H.S. = <math>a_n</math></p> <p><math>= a_{k+1}</math></p> <p><math>= \frac{a_{k-1} - (k-2)a_k}{(k+1)}</math></p> <p><math>= \frac{\frac{k}{(k-1)!} - (k-2)\frac{k+1}{k!}}{(k+1)}</math></p> <p><math>= \frac{k^2 - (k^2 - k - 2)}{k!(k+1)}</math></p> <p><math>= \frac{k+2}{(k+1)!}</math></p> <p><math>= \frac{n+1}{n!}</math> where <math>n=k+1</math></p> | <p>✓</p> <p>✓</p> <p>✓</p> | <p>✓</p> <p><math>(n+1)a_{n+1} = a_{n-1} - (n-2)a_n</math></p> <p>If the statement true for <math>n=k</math>, then proved true for <math>n=k+1</math>. Since true for <math>n=1, 2</math>, <math>\therefore</math> true for <math>n=3, 4, 5, \dots</math> <math>\therefore</math> true for all <math>n \geq 1</math>.</p> |          |

- |        |        |
|--------|--------|
| 1. WAM | 5. WAM |
| 2. MAX | 6. WAM |
| 3. HAR | 7. MAX |
| 4. DIN | 8. MAX |

2009 Year 12 Mathematics Extension 2 Task 4 Trial HSC SOLUTIONS

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Comments                                                                                                                    | Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Comments                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>6 b) i) <math>y = 8x^2 - x^4</math></p> $x^4 - 8x^2 = -y$ $(x^2 - 4)^2 = 16 - y$ $x^2 = \pm \sqrt{\pm \sqrt{16 - y} + 4}$ <p>let <math>x_1</math> and <math>x_2</math> be endpoints of <math>l</math> with <math>0 \leq x_1 \leq x_2</math></p> $\therefore x_2 = +\sqrt{+\sqrt{16 - y} + 4}$ $x_1 = -\sqrt{-\sqrt{16 - y} + 4}$ $\therefore \text{Area} = \pi (x_2^2 - x_1^2)$ $= \pi (\sqrt{16 - y} + 4 - (-\sqrt{16 - y} + 4))$ $= 2\pi \sqrt{16 - y}$ <p>ii) <math>V = \int_0^{16} 2\pi \sqrt{16 - y} \, dy</math></p> $= \left[ -\frac{4\pi}{3} (16 - y)^{\frac{3}{2}} \right]_0^{16}$ $= \frac{4^4 \pi}{3} \quad \checkmark$ | <p><math>\checkmark</math></p> <p><math>\checkmark</math></p> <p><math>\checkmark</math></p> <p><math>\checkmark</math></p> | <p>c) i) <math>y^2 - x^2 = 1</math></p> $\therefore 2y \frac{dy}{dx} - 2x = 0$ $\therefore \frac{dy}{dx} = \frac{x}{y}$ <p>at <math>P(r, \sqrt{1+r^2})</math></p> $\therefore m_{\text{Normal}} = -\frac{\sqrt{1+r^2}}{r}$ $\therefore \tan \theta = \frac{\sqrt{1+r^2}}{r}$ <p>ii) Forces on P</p>  <p>Resolving Vertically and Horizontally</p> $N \sin \theta = mg \quad \text{--- (1)}$ $N \cos \theta = m r \omega^2 \quad \text{--- (2)}$ <p><math>\textcircled{1} \div \textcircled{2} \quad \tan \theta = \frac{g}{r \omega^2}</math></p> $\therefore \frac{\sqrt{1+r^2}}{r} = \frac{g}{r \omega^2}$ $\therefore r = \frac{\sqrt{g^2 - \omega^4}}{\omega^2} \quad \checkmark$ <p>From <math>\textcircled{2}</math>: <math>N^2 = m^2 r^2 \omega^4 \sec^2 \theta</math></p> $= m^2 \omega^4 (r^2 + r^2 \tan^2 \theta)$ $= m^2 \omega^4 \left( \frac{g^2}{\omega^4} - 1 + \frac{g^2}{\omega^4} \right)$ $= m^2 (2g^2 - \omega^4)$ $\therefore N = \sqrt{2g^2 - \omega^4} \quad \checkmark$ <p>iii) <math>\omega^4 \leq g^2 \quad \therefore \omega \leq \sqrt{g} \quad \checkmark</math></p> | <p><math>\checkmark</math></p> <p><math>\checkmark</math></p> <p><math>\checkmark</math></p> <p><math>\checkmark</math></p> <p><math>\checkmark</math></p> <p><math>\checkmark</math></p> <p><math>\checkmark</math></p> <p><math>\checkmark</math></p> |

$y \leq 5 \quad \therefore \sqrt{1+r^2} \leq 5 \quad \therefore \frac{g}{\omega^2} \leq 5 \quad \checkmark$

$\therefore \sqrt{\frac{g}{5}} \leq \omega \leq \sqrt{g} \quad \checkmark$

## 2009 Year 12 Mathematics Extension 2 Task 4 Trial HSC SOLUTIONS

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Comments | Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Comments |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <p><u>Question 7</u></p> <p>a) i) For upward motion<br/> <math>x \uparrow \quad mg \downarrow \quad R \downarrow</math><br/> <math>F = m\ddot{x} = -mg - Rv</math><br/> <math>\therefore \ddot{x} = -(g + kv)</math><br/> <math>v \frac{dv}{dx} = -(g + kv)</math><br/> <math>\frac{dx}{dv} = \frac{-v}{g + kv}</math> ✓<br/> <math>k \frac{dx}{dv} = -1 + \frac{g}{g + kv}</math><br/> <math>\therefore \frac{k^2}{g} \frac{dx}{dv} = -\frac{k}{g} + \frac{k}{g + kv}</math><br/> <math>\therefore \frac{k^2}{g} x = -\frac{k}{g} v + \ln(g + kv) + C</math><br/> at <math>x = 0, v = V</math><br/> <math>\therefore C = \frac{k}{g} V - \ln(g + kV)</math><br/> <math>\therefore \frac{k^2}{g} x = \frac{k}{g} (V - v) - \ln\left(\frac{g + kV}{g + kv}\right)</math><br/> or equivalent ✓<br/> at <math>x = h, v = 0</math><br/> <math>\therefore h = \frac{g}{k^2} \left( \frac{k}{g} V - \ln\left(1 + \frac{k}{g} V\right) \right)</math> ✓</p> |          | <p>ii) <math>\ddot{x} = -(g + kv)</math><br/> <math>\frac{dv}{dt} = -(g + kv)</math><br/> <math>\therefore \frac{dt}{dv} = -\frac{1}{g + kv}</math> ✓<br/> <math>t = -\frac{1}{k} \ln(g + kv) + C</math><br/> at <math>t = 0, v = V</math><br/> <math>\therefore t = \frac{1}{k} \ln\left(\frac{g + kV}{g + kv}\right)</math><br/> or equivalent<br/> at <math>v = 0, t = T</math><br/> <math>\therefore T = \frac{1}{k} \ln\left(1 + \frac{k}{g} V\right)</math><br/> or equivalent ✓</p> <p>iii) Downward motion.<br/> <math>\downarrow mg \quad \uparrow R</math><br/> <math>\therefore F = m\ddot{x} = mg - mkv</math><br/> <math>\frac{dv}{dt} = g - kv</math><br/> <math>\frac{dt}{dv} = \frac{1}{g - kv}</math><br/> <math>t = -\frac{1}{k} \ln(g - kv) + C</math><br/> at <math>t = 0, v = 0 \therefore C = \frac{1}{k} \ln g</math> ✓</p> |          |

## 2009 Year 12 Mathematics Extension 2 Task 4 Trial HSC SOLUTIONS

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Comments | Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Comments |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <p>part iii) continued</p> $\therefore t = \frac{1}{k} \ln\left(\frac{g}{g-kv}\right)$ <p>or <math>-kt = \ln\left(\frac{g-kv}{g}\right)</math></p> $\frac{g-kv}{g} = e^{-kt}$ $1 - \frac{k}{g}v = e^{-kt}$ $\therefore v = \frac{g}{k}(1 - e^{-kt})$ <p>at <math>t \rightarrow \infty, e^{-kt} \rightarrow 0</math></p> $v \rightarrow \frac{g}{k} \text{ (Terminal } v) \checkmark$ <p>iv) If <math>V = \frac{2g}{k}</math></p> $h = \frac{g}{k^2}(2 - \ln 3) \text{ from part i)}$ <p>and sub in <math>-\frac{k}{g}y = \frac{k}{g}vt + \ln\left(1 - \frac{k}{g}v\right)</math></p> $\therefore \frac{k}{g} \cdot \frac{g}{k^2}(2 - \ln 3) = \frac{k}{g}v + \ln\left(1 - \frac{k}{g}v\right)$ $\therefore \frac{kv}{g} + 2 - \ln 3 + \ln\left(1 - \frac{kv}{g}\right) = 0 \checkmark$ |          | <p>let <math>u = \frac{kv}{g}</math></p> $\therefore f(u) = u + 2 - \ln 3 + \ln(1-u)$ $f'(u) = 1 - \frac{1}{1-u}$ <p>at <math>u_1 = 0.81</math></p> $u_2 = 0.81 - \frac{0.81 + 2 - \ln 3 + \ln(0.19)}{1 - \frac{1}{0.19}}$ $\doteq 0.82 \text{ (2 d.p.)} \checkmark$ $\therefore \frac{kv}{g} \doteq 0.82$ <p>and <math>v \doteq 0.82 \frac{g}{k}</math></p> <p>or 82% of terminal velocity on return <math>\checkmark</math></p> <p>v) from part iii)</p> $kt = -\ln\left(1 - \frac{kv}{g}\right)$ <p>at <math>\frac{kv}{g} = 0.82, kT = -\ln(1 - 0.82)</math></p> $= -\ln(0.18)$ <p>from part ii)</p> $V = \frac{2g}{k} \Rightarrow kT = \ln 3$ $\therefore \frac{-\ln(0.18)}{\ln 3} \doteq 0.64 \checkmark$ |          |

## 2009 Year 12 Mathematics Extension 2 Task 4 Trial HSC SOLUTIONS

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Comments | Suggested Solution (s)                                                                                                                                                                                                       | Comments |
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| <p>7 (b) i)</p> $\frac{P_{k+1}}{P_k} = \frac{\frac{(2n)!}{(k+1)!(2n-k-1)!} \left(\frac{1}{2}\right)^{k+1}}{\frac{(2n)!}{k!(2n-k)!} \left(\frac{1}{2}\right)^k} \left(\frac{1}{2}\right)^{2n-k-1}$ $= \frac{k!(2n-k)!}{(k+1)!(2n-k-1)!} \frac{\left(\frac{1}{2}\right)^{2n-k}}{\left(\frac{1}{2}\right)^{2n-k-1}}$ $= \frac{2n-k}{k+1} \checkmark$ <p>Solve <math>P_{k+1} &gt; P_k</math> for greatest value of <math>P_k</math></p> <p>ie, <math>2n-k &gt; k+1</math></p> $2k < 2n-1$ $k < n - \frac{1}{2}$ <p><math>\therefore P_{k+1} &gt; P_k</math></p> <p>for <math>k=0, 1, 2, \dots, n-1</math></p> <p>and <math>P_{k+1} &lt; P_k</math></p> <p>for <math>k=n+1, n+2, \dots, 2n</math></p> <p><math>\therefore k=n</math> gives greatest value for <math>P_k</math> and hence most likely.</p> |          | <p>ii) <math>P_n = {}^{2n}C_n \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^{2n-n}</math></p> $= \frac{2n!}{n!n!} \cdot \left(\frac{1}{2}\right)^n \cdot \left(\frac{1}{2}\right)^n$ $= \frac{2n!}{(n!)^2} \checkmark$ |          |

2009 Year 12 Mathematics Extension 2 Task 4 Trial HSC SOLUTIONS

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Comments                                                    | Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Comments |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <p><u>Question 8</u></p> <p>(a)i) Let <math>u = (1 - \ln x)^n</math><br/> <math>\therefore u' = \frac{n(1 - \ln x)^{n-1}}{-x}</math></p> <p><math>\therefore I_n = \left[ x(1 - \ln x)^n \right]_1^e + n \int_1^e (1 - \ln x)^{n-1} dx</math> ✓</p> <p><math>\therefore I_n = -1 + n I_{n-1}</math></p> <p>ii) <math>I_0 = \int_1^e 1 dx</math><br/> <math>= e - 1</math> ✓</p> <p><math>\therefore I_3 = -1 + 3I_2</math><br/> <math>= -1 + 3(-1 + 2I_1)</math><br/> <math>= -4 + 6(-1 + I_0)</math><br/> <math>= -16 + 6e</math> ✓</p> <p>iii) <math>\frac{I_n}{n!} = \frac{-1}{n!} + \frac{n I_{n-1}}{n!}</math><br/> <math>= \frac{-1}{n!} + \frac{I_{n-1}}{(n-1)!}</math> ✓</p> <p><math>\sum_{r=1}^n \frac{I_r}{r!} = -\sum_{r=1}^n \frac{1}{r!} + \sum_{r=1}^n \frac{I_{r-1}}{(r-1)!}</math><br/> <math>= -\sum_{r=1}^n \frac{1}{r!} + \sum_{r=0}^{n-1} \frac{I_r}{r!}</math></p> | <p><math>v' = 1</math><br/> <math>v = x</math></p> <p>✓</p> | <p>✓</p> <p><math>\therefore \frac{I_n}{n!} = -\sum_{r=1}^n \frac{1}{r!} + \frac{I_0}{0!}</math><br/> <math>= -\sum_{r=1}^n \frac{1}{r!} + \frac{-1}{0!} + \frac{e!}{0!}</math><br/> <math>= e - \sum_{r=1}^n \frac{1}{r!}</math></p> <p>✓ reasoning to answer</p> <p>iv) <math>1 \leq x \leq e</math><br/> <math>\therefore 0 &lt; \ln x \leq 1</math><br/> <math>0 \leq (1 - \ln x)^n \leq 1</math></p> <p>so <math>0 \leq \int_1^e (1 - \ln x)^n dx \leq \int_1^e 1 dx</math><br/> <math>0 \leq I_n \leq e - 1</math></p> <p>v) <math>0 \leq \frac{I_n}{n!} \leq \frac{e-1}{n!}</math><br/> as <math>n \rightarrow \infty</math>, <math>\frac{e-1}{n!} = 0</math><br/> <math>\therefore \lim_{n \rightarrow \infty} \frac{I_n}{n!} = 0</math> ✓ and the deduce.</p> |          |

so  $\lim_{n \rightarrow \infty} \left( e - \sum_{r=0}^n \frac{1}{r!} \right) = \lim_{n \rightarrow \infty} \frac{I_n}{n!} = 0$   
 $\therefore e - \lim_{n \rightarrow \infty} \sum_{r=0}^n \left( \frac{1}{r!} \right) = 0$   
 $\lim_{n \rightarrow \infty} \sum_{r=0}^n \left( \frac{1}{r!} \right) = e$

## 2009 Year 12 Mathematics Extension 2 Task 4 Trial HSC SOLUTIONS

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Comments | Suggested Solution (s) | Comments |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------|----------|
| <p>Q8 (b) i)</p> <p>Since <math>a &gt; 0</math>, <math>\int_0^a \{\lambda \phi(x) + \psi(x)\}^2 dx \geq 0</math> for all real <math>\lambda</math> ✓</p> <p><math>\therefore \int_0^a \{\lambda \phi(x) + \psi(x)\}^2 dx = \lambda^2 \int_0^a \{\phi(x)\}^2 dx + 2\lambda \int_0^a \phi(x)\psi(x) dx + \int_0^a \{\psi(x)\}^2 dx</math> ✓</p> <p>Considered as a quadratic in <math>\lambda</math>, <math>\Delta \leq 0</math>.</p> <p><math>\therefore 4 \left\{ \int_0^a \phi(x)\psi(x) dx \right\}^2 - 4 \int_0^a \{\phi(x)\}^2 dx \cdot \int_0^a \{\psi(x)\}^2 dx \leq 0</math> ✓</p> <p><math>\therefore \left\{ \int_0^a \phi(x)\psi(x) dx \right\}^2 \leq \int_0^a \{\phi(x)\}^2 dx \cdot \int_0^a \{\psi(x)\}^2 dx</math> ✓</p> |          |                        |          |
| <p>ii) if <math>a = 1</math> and <math>\psi(x) = 1</math> ✓</p> <p><math>\int_0^1 1^2 dx = 1</math> ✓</p> <p><math>\therefore \left\{ \int_0^1 \phi(x) dx \right\}^2 \leq \int_0^1 \{\phi(x)\}^2 dx</math> ✓</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                        |          |
| <p>iii) <math>\left\{ \int_0^1 \phi(x) dx \right\}^4 \leq \left\{ \int_0^1 \{\phi(x)\}^2 dx \right\}^2 \leq \int_0^1 \left\{ \{\phi(x)\}^2 \right\}^2 dx</math> ✓</p> <p><math>\therefore \left\{ \int_0^1 f(x) dx \right\}^4 \leq \int_0^1 \{f(x)\}^4 dx</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |                        |          |